Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Chicago, IL

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production engineering experience building backend services, REST APIs, and cloud platforms using Java, Python, React, Node.js, AWS, and Kubernetes. Current research applying LLMs to Android automation, with hands-on experience integrating AI capabilities into enterprise-grade applications.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.

Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs, Web Services.

Databases: PostgreSQL, MySQL.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Azure, Google Cloud Platform.

AI & Identity: Gemini, Vertex AI, LangChain, MCP, HashiCorp Vault, IAM, OAuth-based access control.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Built and maintained backend automation services in Python (Boto3) integrating with AWS APIs, replacing legacy Ruby scripts and reducing deployment time by **40%** across enterprise environments.
- Executed **44 production environment upgrades** in 4 months, debugging and resolving issues across development, test, and production across Linux, EC2, databases, and Teamcenter X services.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, and HashiCorp Vault, strengthening identity and secrets management across the platform.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** and supporting operational excellence initiatives.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in enterprise-grade decision workflows.
- Designed **17-tool MCP server** exposing structured LLM tool-calling APIs for data extraction and cross-system comparison, consumable by external AI agents and enterprise integrations.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).